

Respiration in Plants

1. **Cellular Respiration:** Cellular respiration is defined as the oxidation of food materials (breaking of C-C bonds of complex molecules like $C_6H_{12}O_6$) that occurs within the cell to release energy for ATP synthesis. The energy released is used to synthesize ATP, which acts as the energy currency of the cell.

2. **Respiratory Substrates:** Compounds that are oxidized during respiration are called respiratory substrates. Examples include Carbohydrates (like Glucose, the most common), proteins, fats, and organic acids. The energy source for respiration ultimately derives from photosynthesis.

3. **Respiration Strategy:** Although complete combustion of glucose releases energy, glucose is oxidized in several small steps. This multi-step oxidation enables coupling of the released energy to ATP synthesis.

4. **Gas Exchange in Plants (Breathing):** For respiration, plants take in O_2 and give out CO_2 . Gas exchange occurs via stomata and lenticels. Plants do not require specialized respiratory organs because each plant part takes care of its own gas-exchange needs, and gas transport is very limited. Parenchyma cells in leaves, stems, and roots are loosely packed, providing interconnected air spaces.

II. Glycolysis (EMP Pathway)

1. **Definition and Location:** Glycolysis, also known as the EMP pathway (named after Gustav Embden, Otto Meyerhof, and J. Parnas), is the partial oxidation (breakdown) of glucose to

two molecules of pyruvic acid (pyruvate, a 3-Carbon molecule).

2. Oxygen Requirement: Glycolysis occurs in the absence of O_2 .

It occurs universally in the cytoplasm of all living organisms.

3. Initiation in Plants: In plants, glucose is derived either from the end product of photosynthesis (sucrose) or from storage carbohydrates. Sucrose is converted into glucose and fructose by the enzyme invertase, and these monosaccharides then enter the glycolytic pathway.

4. Key Energy Steps (10 steps total):

- **ATP Utilization (Loss):** 2 ATP molecules are utilized (lost). This occurs during the conversion of glucose into glucose 6-phosphate and the conversion of fructose 6-phosphate to fructose 1, 6-bisphosphate.
- **Redox Reaction:** The product PGAL (3-phosphoglyceraldehyde) is oxidized, and two redox-equivalents (2 H-atoms) are removed and transferred to NAD + forming $NADH + H^+$
- **ATP Synthesis (Gain):** 4 ATP molecules are directly synthesized. Energy is trapped by ATP formation when 1,3-bisphosphoglycerate (BPGA) becomes 3-phosphoglyceric acid (PGA) and when Phosphoenol pyruvate (PEP) converts to pyruvic acid.

5. Net Outcome (per glucose): Net gain is 2 ATP and 2 NADH.

III. Fate of Pyruvate and Aerobic/Anaerobic Pathways

1. Three Fates of Pyruvic Acid: In different cells, pyruvic acid is handled in three ways, depending on the cellular need and oxygen availability:

- Lactic acid fermentation (Anaerobic)
- Alcoholic fermentation (Anaerobic)
- Aerobic respiration (Krebs' cycle)

2. Anaerobic Respiration (Fermentation):

- It is the incomplete oxidation of glucose under anaerobic conditions. This results in limited energy production, releasing less than 7% of the energy in glucose, yielding only 2 ATP. Hazardous products (acid or alcohol) are formed.

- Alcoholic Fermentation: Pyruvic acid is converted to CO_2 and ethanol in organisms like Yeast. The enzymes involved are pyruvic acid decarboxylase and alcohol dehydrogenase. Yeasts are harmed when alcohol concentration reaches about 13%.

- Lactic Acid Fermentation: Pyruvic acid is converted to lactic acid. This occurs in some bacteria and in animal muscle cells when oxygen is inadequate during exercise, catalyzed by lactate dehydrogenase.

- Note: In both fermentation types, the reducing agent $\text{NADH} + \text{H}^+$ is reoxidised to NAD^+ . Aerobic Respiration & Link Reaction (Oxidative Decarboxylation):

- Aerobic respiration requires the presence of oxygen and involves the complete oxidation of organic substances, yielding CO_2 and H_2O . The total ATP yield is approximately 38 ATP.

- Pyruvate from the cytoplasm is transported into the mitochondria.

- Pyruvate (3C) enters the mitochondrial matrix and undergoes oxidative decarboxylation catalyzed by pyruvic dehydrogenase.
- This reaction requires coenzymes like NAD⁺ and Coenzyme A.
- Pyruvate is converted to Acetyl CoA (2C) and CO₂, 2 NADH molecules are produced from the two pyruvic acid molecules.

IV. Krebs Cycle (TCA Cycle/Citric Acid Cycle)

1. Location and Discovery: The cycle takes place in the mitochondrial matrix. It is also known as the Tricarboxylic Acid (TCA) Cycle or Citric Acid Cycle, first elucidated by Hans Krebs.

2. Steps (Starting with Acetyl CoA):

- The Acetyl group condenses with oxaloacetic acid (OAA, 4C) and water to form citric acid (6C), catalyzed by citrate synthase enzyme.
- Citrate is isomerized to isocitrate.
- Isocitrate undergoes decarboxylation to α -ketoglutaric acid (5C).
- α -ketoglutaric acid undergoes decarboxylation to succinyl-CoA.
- Succinyl-CoA is converted to succinic acid (Succinate, 4C), and a GTP molecule is synthesized (substrate level phosphorylation). GTP is then converted to ATP from ADP in a coupled reaction.
- Succinate is oxidized to Fumarate, and then to Malate, and finally oxidized to regenerate OAA (4C). Continued oxidation of acetyl CoA requires the continued replenishment

of OAA.

3. Products Per Glucose (2 turns):

- NAD^+ is reduced to $\text{NADH} + \text{H}^+$ at 3 points per turn. Thus, 6 NADH total per glucose.
- FAD^+ is reduced to FADH_2 at 1 point per turn. Thus, 2 FADH_2 total per glucose.
- 2 ATP (or GTP converted to ATP) are produced via substrate level phosphorylation.
- Total CO_2 produced in complete aerobic oxidation of one glucose molecule is 6 (2 during oxidative decarboxylation, 4 during Krebs cycle).

V. Electron Transport System (ETS) and Oxidative Phosphorylation:-

1. ETS Location and Function: The ETS is a metabolic pathway located in the inner mitochondrial membrane. Its primary purpose is to release and utilize the energy stored in $\text{NADH} + \text{H}^+$ and FADH_2 (formed during the preceding cycles) by oxidation. This energy is used to synthesize ATP.
2. Electron Flow: Electrons pass from one carrier to another. Eventually, the electrons are passed onto O_2 to form H_2O . In aerobic respiration, Oxygen acts as the final hydrogen/electron acceptor.

3. The Electron Transport Complexes:

- Complex I (NADH dehydrogenase): Oxidizes electrons from NADH . Contains Flavin Mono Nucleotide (FMN) and FeS. Pumps protons (H^+) into the inter-membrane space.

- Complex II: Deals with FADH_2 . It contains Flavin Adenine Di Nucleotide (FADN) and FeS. (In the diagram, it is associated with succinate dehydrogenase).
- Ubiquinone (Q): Receives electrons/reducing equivalents from Complex I and Complex II. Reduced ubiquinone (Ubiquinol) is then oxidized by Complex III.
- Complex III (Cyt bc₁ complex): Contains Cyt b, FeS, and Cyt c₁. Pumps protons into the inter-membrane space.
- Cytochrome c (Cyt c): A small mobile protein attached to the outer surface of the inner membrane. It acts as a mobile carrier between Complex III and Complex IV.
- Complex IV (Cytochrome c oxidase): Contains cytochromes a and a₃, along with 2 copper centres (CuA and CuB). This complex transfers electrons to the final acceptor (O_2).

4. Oxidative Phosphorylation (ATP Synthesis):

- This process uses the energy from oxidation-reduction reactions for phosphorylation. It is distinct from photophosphorylation, which uses light energy to create the proton gradient.
- ATP Synthase (Complex V): Utilizes the energy released by the ETS to synthesize ATP. It has two major components:
 - F₁ headpiece: The peripheral membrane protein complex that projects into the Matrix. It is the site for ATP synthesis ($\text{ADP} + \text{P}_i \rightarrow \text{ATP}$).
 - F₀: The integral membrane protein complex that forms a channel through which protons cross the inner membrane.

- **Proton Gradient:** The electrochemical proton gradient drives ATP synthesis. For each ATP produced, 2H⁺ pass through F₀ from the inter-membrane space to the matrix.

VI. Respiratory Balance and Amphibolic Pathway

1. **Total NADH and ATP Yield:** During the complete aerobic oxidation of one glucose molecule, 10 NADH molecules are produced (2 from glycolysis, 2 from oxidative decarboxylation, 6 from the Krebs cycle). The theoretical Net Gain of ATP is 38 ATP.

2. **Assumptions for ATP Balance:** This calculation assumes that all steps occur sequentially, NADH from glycolysis is transferred efficiently into the mitochondria for oxidation, intermediates are not withdrawn, and only glucose is respired. In reality, these assumptions are often not valid as pathways work simultaneously, and substrates are withdrawn or enter pathways as needed.

3. **Amphibolic Pathway:** The respiratory pathway is considered an amphibolic pathway because it serves both catabolic (breakdown) and anabolic (synthesis) functions.

- **Breakdown (Catabolism):** Glucose is the favored substrate. Fats are broken down into glycerol (which forms PGAL) and fatty acids (which form acetyl CoA). Proteins are broken down into amino acids, which enter the cycle at the level of pyruvate or acetyl CoA.

- **Synthesis (Anabolism):** When the organism needs to synthesize compounds like fatty acids, intermediates such as acetyl CoA are withdrawn from the respiratory pathway.

VII. Respiratory Quotient (RQ)

1. Definition: RQ is the ratio of the volume of CO_2 evolved to the volume of O_2 consumed in respiration.
2. Formula: $\text{RQ} = \text{Volume of } \text{CO}_2 \text{ evolved} / \text{Volume of } \text{O}_2 \text{ consumed}$.
3. Dependence: RQ depends upon the type of respiratory substrate.

4. RQ Values:

- Carbohydrates: $\text{RQ} = 1.0$. This occurs because equal amounts of CO_2 and O_2 are evolved and consumed (e.g., $6\text{CO}_2/6\text{O}_2$).
- Fats: $\text{RQ} < 1$ (e.g., 0.7 for tripalmitin).
- Proteins: $\text{RQ} = 0.9$.

The overall process of cellular respiration resembles a cellular power grid where glycolysis and the Krebs cycle serve as generating stations producing high-energy carriers (NADH and FADH_2), and the ETS acts as the hydroelectric dam, using the flow of electrons (water) to build up a proton gradient (pressure) that finally drives the ATP synthase turbine to produce the usable energy currency (ATP).